



Effect of graphene oxide on thermal charging and discharging behaviour of paraffin wax as nano-enhanced phase change materials

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Abstract

This study investigates the impact of adding graphene oxide (GO) to paraffin wax as a phase change material (PCM) on its thermal behaviour. GO addition, confirmed visually, enhances thermal conductivity by 60%. Structural features of both GO and the prepared PCM are reported. The investigation considers parameters such as GO concentration and heat transfer fluid flow rate. Increasing GO concentration (0.25–1.00 vol%) decreases PCM's thermal charging and discharging due to improved heat transfer. Moreover, increasing flow rate (0.375–1.5 L/min) reduces charging time. Higher flow rates correspond to improved efficiency in the thermal energy storage system. These findings suggest GO is an effective additive for enhancing paraffin wax as a PCM, beneficial for energy storage, thermal management, and building insulation applications. The study implies practical implications for utilizing GO-infused PCM in diverse thermal management systems, highlighting its potential in enhancing thermal properties and overall system efficiency.

Introduction

The world is witnessing a significant surge in energy demand driven by ongoing industrial expansion and a growing global population. Unfortunately, heavy reliance on fossil fuels as primary energy source has resulted in severe consequences such as climate change, global warming, and ecological damage [1]. As a result, it has become imperative to address these issues by transitioning towards clean and environmentally friendly energy solutions. Renewable energy like geothermal, solar, and wind power have garnered immense attention as sustainable alternatives to fossil fuels, offering the advantage of being eco-friendly and their abundance. Among these options, solar energy stands out due to the exceeding immediate demand. However, seasonal variations of solar energy may restrict its utility. Nevertheless, solar energy remains a highly promising alternative energy source that can maintain the demand and supply circularity [1]. Therefore, to develop and design an efficient solar thermal energy storage system (TESS) is today's need-of-the-hour. TESS is energy storage device in which a

working substance absorbs or emits energy with respect to temperature variations. Majorly, the popular utility of TESS deals with the temperature-controlled atmosphere in various industries such as drug transportation, swimming pools and centrally air-conditioned buildings. Among all of the systems, latent heat storage systems (LHSS) works on principle of energy storage in form latent heat of working substance during isothermal phase change (solid $\rightleftharpoons$ liquid) [2]. LHSS has two major advantages over other systems; (i) high storage density, and (ii) smooth temperature variations [3]. In this view, phase change materials (PCMs) were found to be excellent candidates as working substance to store solar thermal energy such as paraffin wax (organic), salt hydrates (inorganic) and fatty acids (organic) [4]. PCM-based LHSSs have advantages of high energy storage density, chemical stability, constant heat release temperature, and low degree of supercooling [5]. However, low thermal conductivity (TC), easy phase leakage and poor light absorption of traditional organic PCMs hinder its performance by restricting their scalability and commercialization [6]. Therefore, the chemistry of traditional paraffin wax has been modified by mixing other components for enhanced thermal performance of TESSs [7]. Zeng et al. [8] recorded a 356.25% increment in TC of organic PCM (i.e. 1-tetradecanol) by addition of Ag nanowires (62.73 wt%). Li et al. [9] prepared paraffin wax-based composite samples by using montmorillonite and multi-walled carbon nanotubes as reinforcements to understand the TC variation in which ~ 165% enhancement

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was observed as collated to bare paraffin wax. Li et al. [10] studied effect of graphene sheets on thermophysical properties of PCMs (stearic acid and lauric acid, respectively) with increased TC. Liang et al. [11] studied the incorporation of lauric acid and graphene (and its derivatives) in paraffin wax for thermal storage applications. It was observed that addition of reinforcement(s) enhanced the charging (CH) and discharging (DCH) efficiency by > 90% with excellent cyclic stability, good shape stability and good phase change reversibility. Benbrika et al. [12] reported increased thermal CH, DCH and reduced thermal storage capacity (~ 14%) of PCMs (dispersed with graphene nanoplatelets (GNPs)) using horizontal cylinder. Kumar and Cheralathan [13] comprehended the CH and DCH performance of low-capacity energy storage tank filled with dissimilar types of heat transfer fluid (HTF) and 35 spherical capsules containing nano PCM. Kumar et al. [14] performed experimental study on TES by using paraffin wax doped with functionalized graphene (concentration = 0.25–1.0 vol%). As a result, it was observed that CH time was reduced with the increase in graphene concentration. Passaro et al. [15] studied heat exchange process of LHSS using compact finned tube heat exchanger with paraffin wax (A53) doped with GNPs. The experimental setup with optimized number of fins was successfully able to decrease discharge time by 60% to 70%. Dong et al. [16] conducted research on the thermal performance of a tube and shell LHSS tank using different nano-enhanced PCMs, specifically GNPs and single-walled carbon nanotubes (SWCNT). Their results demonstrated that the incorporation of SWCNTs into paraffin wax led to a significant reduction (~ 43%) in melting time and an impressive improvement (~ 147%) in stored latent heat energy storage compared to using pure paraffin alone. Similarly, Fikri et al. [17] investigated the TC, stability, and reliability of PCM doped with functionalized multi-wall carbon nanotubes. They observed that the TC was enhanced by 150.7%, and the PCM remained thermally stable up to 200 °C.

Till date, several papers have been published focusing on thermal CH, DCH and overall efficiency of TESS using variety of materials enhancing the characteristics of paraffin wax. Still, the incorporation of graphene oxide (GO) into paraffin wax is a state of art in this regard. The present study describes systematic data acquisition and analysis of self-designed horizontal cylinder solar reactor for thermal energy storage applications of paraffin wax.

Experimental section

Figure 1a represents schematic flow diagram of undertaken thermal studies. In the experimental setup, cylinder (5.5 kg capacity) filled with prepared PCM-based composites was installed in the circuit and two separate heat cycles were monitored which are (i) thermal charging and (ii) thermal

discharging as shown in Fig. 1a. In thermal charging cycle, HTF (water in present case) was filled in source tank with the help of centrifugal pump and heated through electric heaters till 85 °C was achieved. Thereafter, the heated HTF was allowed to flow through heat exchanger assembly in which the heat was transferred to PCM-based composite samples and stored again in storage tank (1). The cycle was remained open for charging till the complete melting of paraffin wax occurred which was confirmed by achieving ~ 65 °C at ‘outlet’, shown in Fig. 1a. While, discharging cycle involves the flow of cold HTF fluid from different storage tanks (2) to extract heat energy from PCM-based composite and the fluid will be discarded as shown in Fig. 1a. The completion of thermal discharging cycle was determined by achieving 35 °C of heat exchanger assembly at ‘outlet’ as shown in Fig. 1a. Rest of the assumption, materials and discussion on structural features of GO is given in supplementary information.

Results and discussion

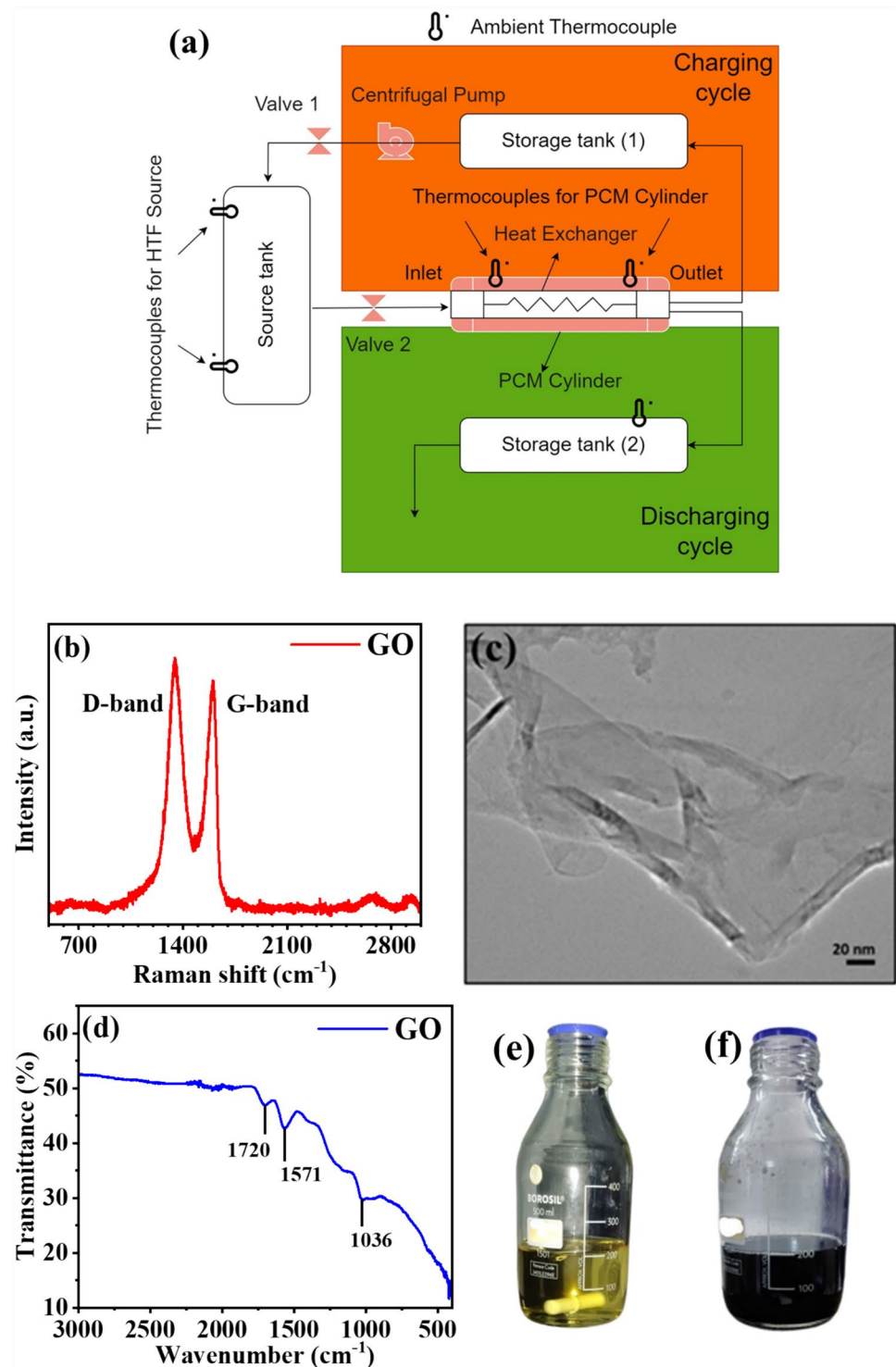
PCM with higher TC can significantly reduce time required for both thermal CH and DCH processes. The effectiveness of any TESS relies heavily on efficiency of these processes. In this study, the exceptional TC properties of GO have been harnessed to enhance TC of paraffin wax, aiming to improve overall thermal performance of system.

Considering the huge increment in TC as a result of GO addition in paraffin wax, the prepared composite samples would be suitable for thermal charging and discharging application in TESSs. Therefore, a separate thermal charging and discharging study has been conducted. Figure 2b shows a detailed graphical representation of temperature profiles and region considered for energy calculation. The process parameter such as GO concentration and flow rate of working fluid plays critical role in executing efficient heat transfer between storage material (prepared samples) and HTF (water). Figure 2a represents the effect of GO concentration (0.25–1.00 vol%) at different HTF flow rates (0.375, 0.75 and 1.5 L/min) on thermal charging and discharging of prepared samples.

Thermal charging

Thermal charging of paraffin wax-based TESS was confirmed by achieving a constant temperature of 65 °C at outlet Fig. 1a. It was quite obvious to record the decreasing overall charging time as an effect of increasing GO concentration. Interestingly, it was observed that overall charging time was reduced upto 0.75 vol% GO (A3) while further increasement in GO concentration (A4) increased the overall charging time. The main reason behind the

Fig. 1 **a** Schematic ray-diagram of experimental setup for thermal studies, **b** Raman spectra, **c** TEM micrograph of procured GO, **d** FTIR spectra, Photographic view of molten **e** paraffin wax, and **f** A4 sample (1.0 vol% GO + paraffin wax)



reduced charging time could be the faster energy transfer from HTF to storage material through additional thermal conductive network within paraffin wax [18]. For A3, about 50.0%, 42.85%, and 43.75% reduction in overall charging time as compared to pure paraffin wax at 0.375, 0.75 and 1.5 L/min flow rate, respectively.

Beyond 0.75 vol% (A4; 1.00 vol%), the overall charging time was found to be increased by 33.33%, 25%, and 22.22% at 0.375, 0.75 and 1.5 L/min flow rates with respect to 0.75 vol% GO (A3). This behaviour can be attributed to the reduced intermolecular free mobility of GO particles at 1.0 vol% concentration of GO (sample A4) [19]. As an

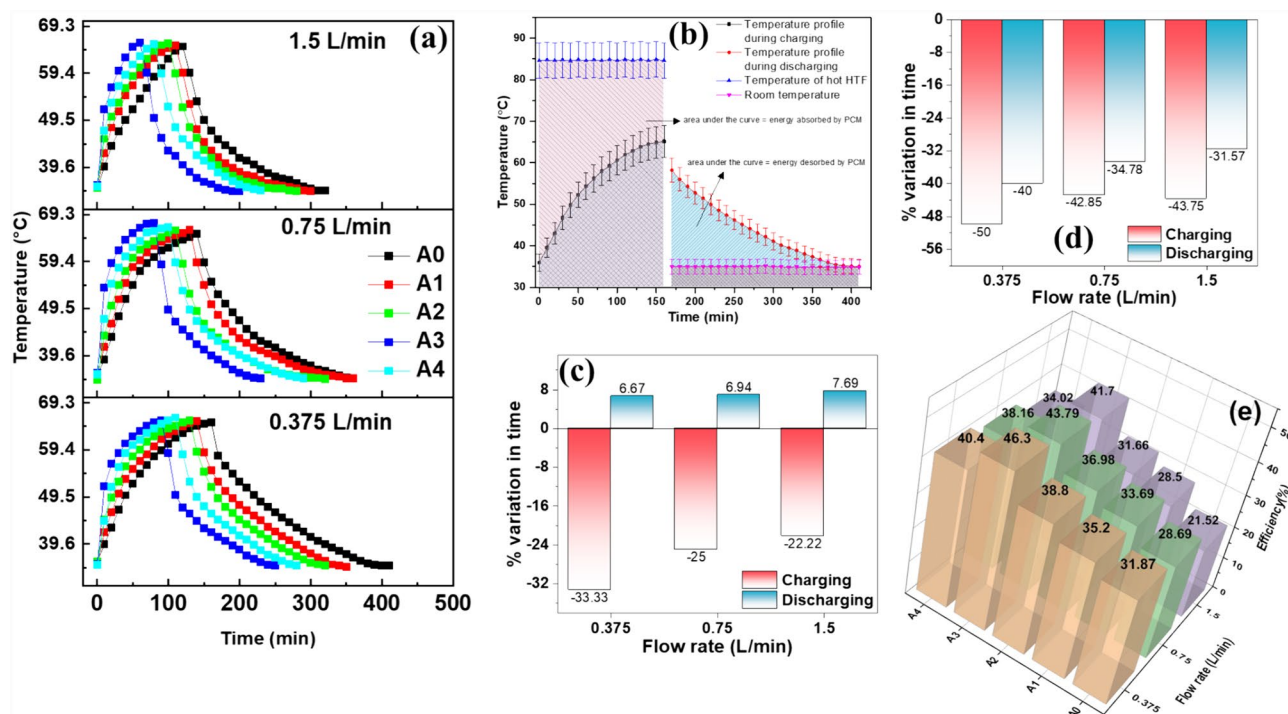


Fig. 2 **a** Temperature profile during charging-discharging of TESS with prepared composite at different flow rates (0.375, 0.75, and 1.5 L/min), **b** Detailed representation of energy during charging and dis-

charging of TESS, **c** A3 (with respect to A0), and **d** A4 (with respect to A3), and **e** Efficiency of adopted TESS at different flow rates (0.375–1.5 L/min) and different energy storage medium (A0–A4)

effect of flow rate, overall charging time has been decreased with increasing flow rate (0.375 to 1.5 L/min) for each GO concentration. The decreased charging time at high flow rate (1.5 L/min) would indicate efficient energy transfer through HTF to working storage material resulting, 25.0% (for A0) and 33.3% (for A3) decrement in overall time (to reach 65 °C) as compared to 0.375 L/min, respectively. An overall charging performance variation of A3 and A4 is also shown in Fig. 2a.

Thermal discharging

Once charging process was completed, discharging of TESS was started so that discharging behaviour of energy storage material, PCM in the present study, can be monitored through heat transfer from PCM to HTF. During discharging process, HTF absorbed the heat from PCM cylinder along with solidification (due to cooling) of paraffin wax or wax-based composite as shown in Fig. 2b. In the present study, normal water (at room temperature) has been used as HTF fluid, circulated through PCM cylinder until temperature of PCM cylinder was reduced to ambient room temperature, i.e., 35 °C (completion of discharging). Effect of GO concentration (A0–A4) on overall discharging time was also studied at different flow rates as shown in Fig. 2a. As an effect of incorporation of GO in paraffin wax, it was clearly

observed that discharging time was reduced with increasing GO concentration. The observed thermal behaviour is completely associated to higher TC of GO as compared to paraffin wax resulting to enhanced heat transfer. As the GO concentration was increased, the rate of heat transfer has also been increased, resulting to a rapid discharging process. For A3, about 40.0%, 34.78%, and 31.57% reduction in overall discharging time was recorded as compared to pure paraffin wax at 0.375, 0.75 and 1.5 L/min flow rate, respectively. Beyond 0.75 vol% (A4; 1.00 vol%), the overall discharging time was found to be increased by 6.67%, 6.94%, and 7.69% at 0.375, 0.75 and 1.5 L/min flow rates with respect to 0.75 vol% GO (A3). At 0.375 L/min, the convective heat transfer allows for faster heat dissipation from the composite, storage material, to the HTF. This accelerates the cooling process, leading to even more rapid reduction in temperature at faster flow rates (0.75 and 1.5 L/min). So, considering specific application requirements, determining the optimal flow rate for discharging of cylinder requires system design and composite characteristics. An overall charging-discharging performance variation of A3 and A4 is shown in Fig. 2c and d, respectively.

Table 1 Different nano materials used in charging and discharging analysis

Year	Modifications	Concentration (wt%/vol%)	Charging time (Min)	Discharging time (Min)	Enhanced thermal efficiency (%)	References
2021	Functionalized Graphene	1	120	–	–	[14]
2021	MWCNT	1	125	150	71	[20]
2021	SWCNT	3	300	540	37	[21]
2021	CuO	1	105	85	17.3	[22]
2023	GO	0.75	60	180	46.3	Present study

Thermal efficiency

Figure 2e represents the overall thermal efficiency (provided in supplementary information) of GO-reinforced paraffin wax as energy storage medium. Here, it has been observed that reduced CH and DCH time, as an effect of GO addition, affected the overall efficiency of the system. At a particular flow rate of HTF, addition of GO (A3; 0.75 vol%) enhanced overall thermal efficiency of PCM-based TESS by ~ 14% as compared to paraffin wax-based TESS. Importantly, sample A4 exhibited reduced thermal efficiency of TESS as compared to A3 which is similar to charging and discharging behaviour. While, the overall efficiency has been reduced with increase in flow rate of HTF irrespective of composition of energy storage medium. In order to achieve maximum efficiency of the TESS, the concentration of GO should be 0.75 vol% in paraffin wax for energy storage medium and the HTF flow rate can be taken as low as possible, i.e., 0.375 L/min. Numerous investigations have delved into the application of different nanomaterials for analyzing charging, discharging processes and enhanced thermal efficiency, highlighting varied concentrations as detailed in Table 1.

Conclusion

Experimental study was successfully conducted on TESS utilizing GO-based paraffin wax composite as energy storage medium. To analyse the effect of flow rate of HTF on CH and DCH, various experiments were performed with different flow rates of 0.375, 0.75 and 1.5 L/min. Following observations have been obtained from this study:

- i. As an effect of GO concentration, 0.75 vol% (A3) exhibited the optimized charging among all the sample with 60 min as overall charging time. Beyond 0.75 vol%, TESS took more time for complete charging. Similarly, A3 exhibited fastest discharging of TESS which is due to reduced intermolecular free mobility of GO particles.
- ii. HTF flow rate greatly affects heat transfer. Higher flow rate leads to shorter charging and discharging times, while lower flow rates result in longer times.
- iii. For A3, ~ 50.0%, 42.85%, and 43.75% reduction in overall charging time has been recorded as compared to pure paraffin wax (A0) at 0.375, 0.75 and 1.5 L/min flow rate, respectively.
- iv. Till 0.75 vol% GO (A3), the overall discharging time was found to be decreased with increasing flow rate. 40.0%, 34.78%, and 31.57% reduction in overall discharging time was recorded as compared to pure paraffin wax (A0) at 0.375, 0.75 and 1.5 L/min flow rate, respectively. While, A4 exhibited longer charging and discharging time as compared to A3, i.e., 33.3% and 6.67%, respectively (at 0.375 L/min).
- v. The highest overall efficiency of 46.3% has been obtained when 0.75 vol% GO-reinforced paraffin wax was used as energy storage medium at 0.375 L/min flow rate.
- vi. Experimental findings indicate that the optimal concentration of GO is 0.75%, enhancing the TC of paraffin wax by 60%.
- vii. To optimize the storage system cost, it is advisable to incorporate 0.75% GO into paraffin wax, resulting in reduced melting time and elevated PCM temperatures.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1557/s43580-024-00895-0>.

Author contributions AK conceptualized and conducted the experimental work, data analysis and wrote manuscript, while AG and KS reviewed the work and edited the manuscript. MS performed the experimental work and data analysis.

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Data availability The data presented in this study are derived solely from our own research. No external reference data were utilized; rather, the data are analyzed and compared within the scope of our work.

Declarations

Competing interests The authors declare that they have no competing interests related to the publication of this research article.

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